

SBN Explorer

Fitness and Reward Functions

Let $F : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$ denote the Boolean transformation implemented by a BC. In the evolutionary framework, *fitness functions* provide global adversarial evaluation criteria, whereas *reward functions* provide local structural signals used to guide mutation and exploration.

1 Fitness Functions

We evaluate the cryptographic quality of F through three standard criteria related to differential, linear, and algebraic cryptanalysis. Each criterion defines a fitness function to be maximized.

Differential Fitness (F1) For $\alpha \in \mathbb{F}_2^n$ and $\beta \in \mathbb{F}_2^m$, the differential distribution table is $\text{DDT}_F(\alpha, \beta) = |\{x \in \mathbb{F}_2^n : F(x) \oplus F(x \oplus \alpha) = \beta\}|$.

The maximum differential probability is

$$\text{DP}_{\max}(F) = \max_{\alpha \neq 0, \beta} \frac{\text{DDT}_F(\alpha, \beta)}{2^n}.$$

A cryptographically strong function is expected to satisfy $\text{DP}_{\max}(F) \leq \varepsilon_D$ for some small threshold ε_D . We define $\text{Fit}_{\text{diff}}(F) = -\log_2(\text{DP}_{\max}(F))$.

Maximizing Fit_{diff} amounts to minimizing the success probability of the best differential characteristic.

Linear Fitness (F2) For $a \in \mathbb{F}_2^n$ and $b \in \mathbb{F}_2^m$, define the correlation

$$C_F(a, b) = \frac{1}{2^n} \sum_{x \in \mathbb{F}_2^n} (-1)^{a \cdot x \oplus b \cdot F(x)}.$$

The maximum linear bias is $\text{LP}_{\max}(F) = \max_{a \neq 0, b \neq 0} |C_F(a, b)|$. Equivalently, in Walsh form,

$$\max_{a \neq 0, b \neq 0} \left| \sum_{x \in \mathbb{F}_2^n} (-1)^{a \cdot x \oplus b \cdot F(x)} \right| \leq \varepsilon_L 2^n,$$

for some small threshold ε_L .

We define $\text{Fit}_{\text{lin}}(F) = -\log_2(\text{LP}_{\max}(F))$.

Maximizing Fit_{lin} reduces the strength of the best linear approximation.

Algebraic Fitness (F3) Algebraic cryptanalysis exploits low-degree polynomial relations between input and output variables. A strong Boolean transformation should therefore exhibit high algebraic complexity.

Let F_i denote the i -th coordinate function of F , written in Algebraic Normal Form (ANF):

$$F_i(x) = \bigoplus_{u \in \mathbb{F}_2^n} a_{i,u} x^u.$$

The algebraic degree of F is $\deg(F) = \max_{1 \leq i \leq m} \deg(F_i)$.

To capture the growth of algebraic complexity under iteration, we consider

$$\deg_k(F) = \deg(F^{(k)}).$$

The algebraic fitness is then defined by $\text{Fit}_{\text{alg}}(F) = \deg_k(F)$.

Maximizing Fit_{alg} favors transformations whose algebraic complexity grows rapidly under composition, providing a practical proxy for resistance to algebraic attacks.

2 Reward Functions

Fitness functions are global and adversarial, but often expensive and locally flat. We therefore complement them with reward functions based on structural variations between a circuit F and a mutated circuit F' . Each reward is defined as a differential indicator $R_i(F \rightarrow F')$.

Effective Algebraic Degree Reward (R1) Let $\deg_k(F) = \max_{1 \leq i \leq m} \deg(F_i^{(k)})$. We define $R_1(F \rightarrow F') = \deg_k(F') - \deg_k(F)$.

R1 promotes mutations increasing multi-round algebraic complexity.

Algebraic Normal Form Entropy (R2) For each coordinate F_i , let $M_i = \{u \in \mathbb{F}_2^n : a_{i,u} = 1\}$ be the set of monomials appearing in its ANF. Define $H_{\text{ANF}}(F) = \frac{1}{m} \sum_{i=1}^m \frac{\log_2(|M_i|+1)}{n}$. Then $R_2(F \rightarrow F') = H_{\text{ANF}}(F') - H_{\text{ANF}}(F)$.

This reward promotes dense and less factorizable ANF representations.

Walsh Spectrum Flattening (R3) For each coordinate F_i , define the Walsh transform $W_{F_i}(a) = \sum_{x \in \mathbb{F}_2^n} (-1)^{F_i(x) \oplus a \cdot x}$.

Let $S(F) = \frac{1}{m} \sum_{i=1}^m \text{Var}_{a \neq 0}(W_{F_i}(a))$.

Then $R_3(F \rightarrow F') = S(F) - S(F')$.

Reducing spectral dispersion discourages dominant linear correlations.

Local Differential Uniformization (R4) Define $\text{DDT}_F(\alpha, \beta) = |\{x : F(x) \oplus F(x \oplus \alpha) = \beta\}|$, and $D(F) = \text{Var}_{\alpha \neq 0, \beta} \left(\frac{\text{DDT}_F(\alpha, \beta)}{2^n} \right)$.

Then $R_4(F \rightarrow F') = D(F) - D(F')$.

This reward penalizes uneven differential behavior and dominant differential transitions.

Dependency Graph Expansion (R5) Let $M(F)$ be the dependency matrix

$$M_{i,j}(F) = \begin{cases} 1 & \text{if } F_i \text{ depends on } x_j, \\ 0 & \text{otherwise.} \end{cases}$$

Let $\rho(F) = \text{rank}(M(F))$.

Then $R_5(F \rightarrow F') = \rho(F') - \rho(F)$.

This reward promotes stronger input-output entanglement.

Symmetry Breaking Reward (R6) Let $\text{Aut}(F)$ denote the automorphism group of the circuit structure, and define $\sigma(F) = |\text{Aut}(F)|$.

Then $R_6(F \rightarrow F') = \sigma(F) - \sigma(F')$.

This reward penalizes structural symmetries and repeated exploitable patterns.